

Finite State Machines

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Agenda

- ◆ Finite State Machine
- ◆ Mealy Machine
- ◆ Moore Machine
- ◆ Design Examples
- ◆ Glitches and Hazards

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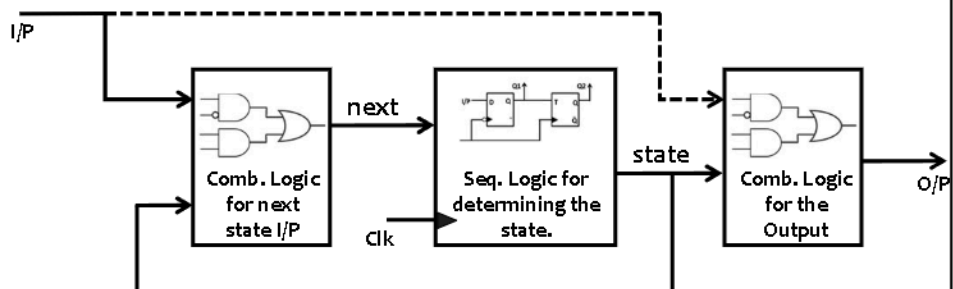
Finite State Machines

- ♦ Systematic way of specifying any sequential logic
- ♦ Ideally suited for complex sequential logic.
- ♦ To specify the sequential circuits which are :
 - ♦ complex in their transition sequence &
 - ♦ depend on several control inputs
- ♦ Different models in FSM are :
 - ♦ Mealy Model
 - ♦ Moore Model

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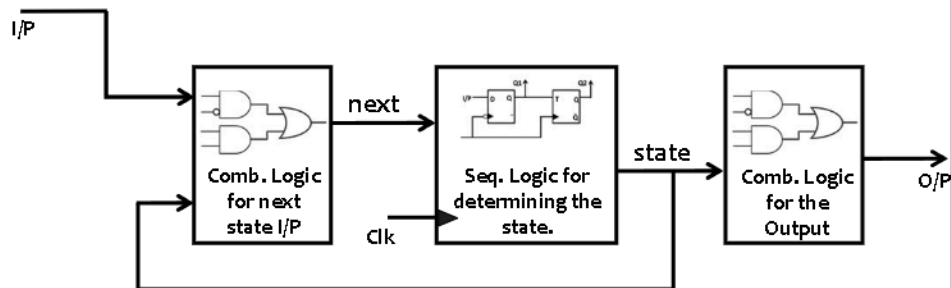
Mealy Machines



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Moore Machines



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Mealy Vs Moore

Mealy

- ♦ Output depends on both input & current state of the circuit.
- ♦ Next state is dependent on both input & current state.
- ♦ Outputs are Asynchronous. So, glitches can occur.
- ♦ Less no. of states. Obviously, less no. of flipflops are sufficient for implementation.
- ♦ Not safe to use.
- ♦ Area efficient.

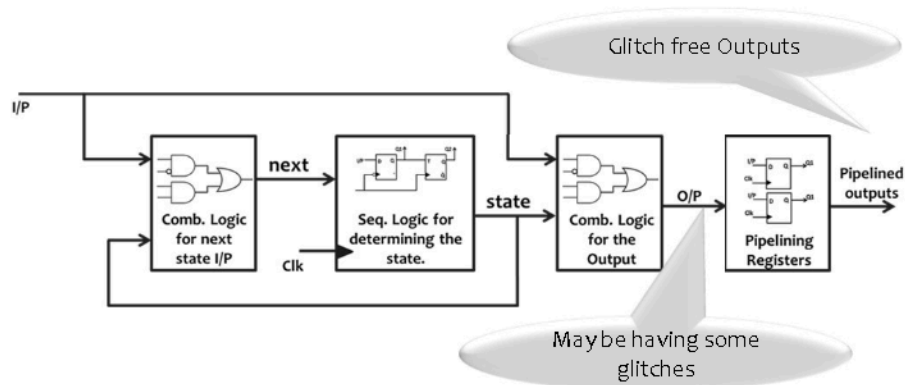
Moore

- ♦ Output depends only on current state of the circuit.
- ♦ Next state is dependent on both input & current state.
- ♦ Outputs are Synchronous. So, glitches can never occur.
- ♦ More no. of states. Obviously, more no. of flipflops are required for implementation.
- ♦ Safe to use.
- ♦ Not area efficient.

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Mealy FSM



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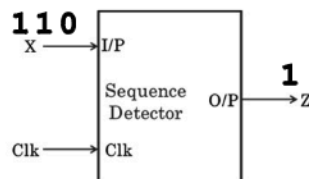
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Sequence Detector - FSM Example

Specification

Design a sequence detector that receives binary data stream at its input, **X** and signals when a combination '110' arrives at the input by making its output **Z** high which otherwise remains low.

Consider the first received bit is LSB



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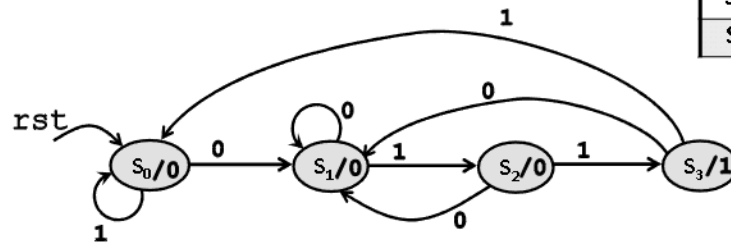
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Sequence Detector - Moore

Sequence : 110

State	Meaning
S_0	Initial state
S_1	0
S_2	10
S_3	110

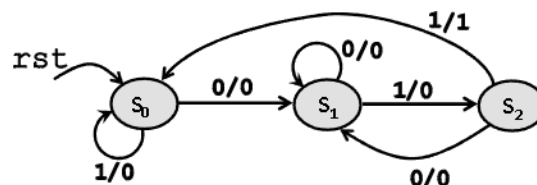


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Sequence Detector - Mealy

Sequence : 110

State	Meaning
S_0	Initial state
S_1	0
S_2	10



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Sequence Detectors

Example

- ♦ Lets consider a sequence recognizer which is capable of detecting 1001 Sequence.

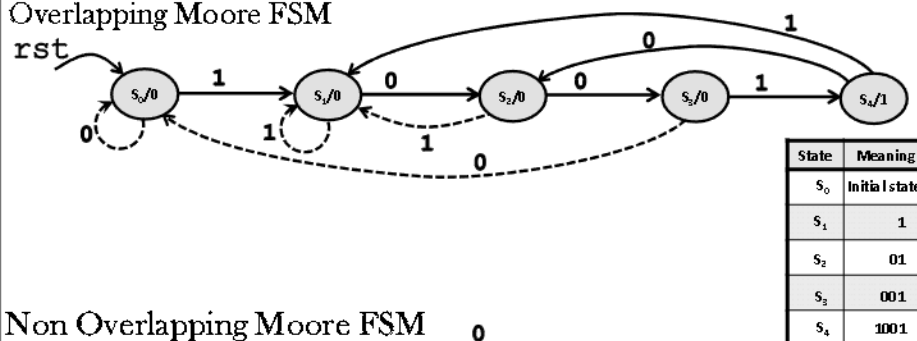
Cycle Number	1	2	3	4	5	6	7	8	9	10	11	12
Input Sequence	0	1	0	0	1	0	0	1	1	0	0	1
O/P for overlapping Seq. Detector	0	0	0	0	1	0	0	1	0	0	0	1
O/P of non overlapping Seq. Detector	0	0	0	0	1	0	0	0	0	0	0	1

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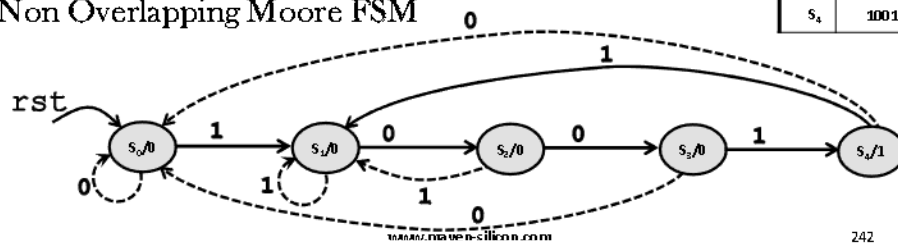
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Sequence Detector for 1001 Sequence

Overlapping Moore FSM



Non Overlapping Moore FSM



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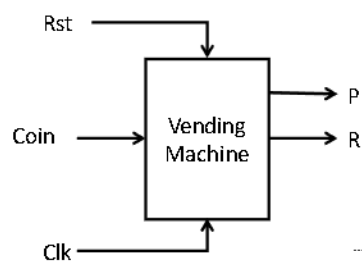
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Vending Machine - FSM Design

Specification

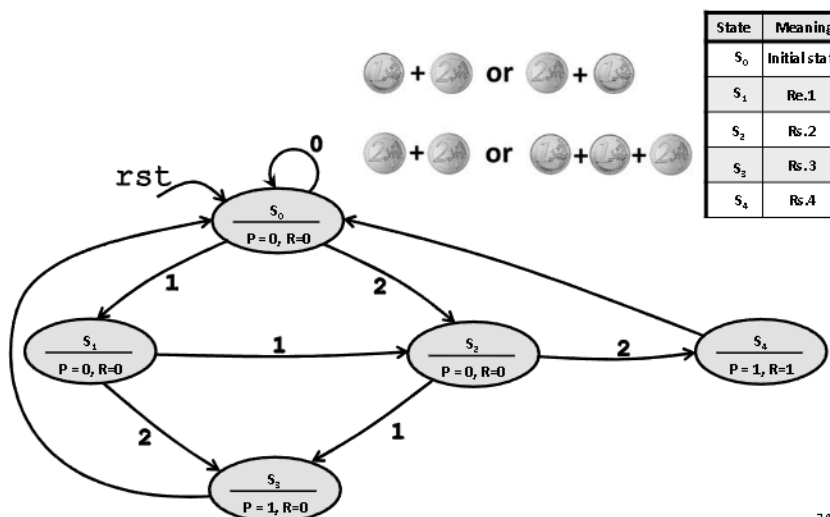
- Design a vending machine which delivers a product when it receives Rs.3/- and also it has to return the change if receives Rs.4/- . The machine can take only two types of coins of denomination 1 and 2 in any order.



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Vending Machine



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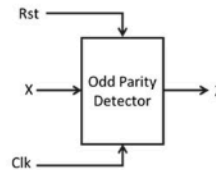
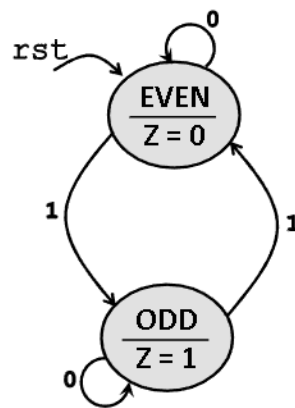
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Odd Parity Detector-FSM Design

Specification

Assert output whenever input bit stream has odd number of 1's.



Present State (Q)	I/P (X)	Next State (Q+)	O/P (Z)
Even	0	Even	0
Even	1	Odd	0
Odd	0	Odd	1
Odd	1	Even	1

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Odd Parity Detector-FSM Design

Specification

Assert output whenever input bit stream has odd number of 1's.

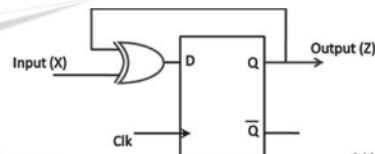
Present State (Q)	I/P (X)	Next State (Q+)	O/P (Z)
Even	0	Even	0
Even	1	Odd	0
Odd	0	Odd	1
Odd	1	Even	1

State Encoding

Design a circuit from FSM will be discussed in advanced course

$$\text{NextState} = \text{Din} = Q \oplus X$$

$$\text{Output (Z)} = Q$$



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